
ONE BIT ACROSS THE WALL: ODD-SUPPLY SEPARATION AND THE ONE-CUT HYPOTHESIS ACROSS TWENTY-THREE ROWS

Block structure, the coordinate bridge as wall, and a halt-locus typing of the barrier literature

RESEARCH ARTICLE (STRUCTURAL ANALYSIS)

To test one structural hypothesis against the dated corpus of twenty-three problem rows, twenty-two open and one a crossed control, claiming a solution for none of them.

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ABSTRACT. Mathematics now carries a growing inventory of proved method-class barriers (relativization, natural proofs, algebrization, the sieve parity problem, optimality and occurrence refutations, independence results), each a theorem about a technique family; the inventory has no common anatomy, and this paper supplies one at the level of structure. Working inside the odd-supply frame of the offering-bit program, in which a problem row is a finite data set with a group action, a reading class, a Boolean target, and a dated corpus, we show that every mapped barrier is an evenness statement: the blocked register is blind to the row's odd target, and exactly one binary orientation coordinate is missing. Across twenty-three rows, the seven Millennium problems and sixteen further open problems, plus ten dated historical crossings, the block shape is exact wherever the corpus holds a Boolean decision bit and a named even register, and the deficit is strictly one bit wide on every row, measured at the row's decision seat under the seat-uniqueness premise of Section 4.1: continuous-carrier rows are wall rows under the orientation-after-construction law, register-less rows are pure unbridged targets, and undecidability-flavored rows sit at the diagonal port. The hypothesis is falsifiable and armed: a provenance-clean height pairing nondegenerate at analytic order two for elliptic curves is the framework's named next deed, and every future crossing is predicted to enter through a certificate the prior even register could not denote. If confirmed, the barrier literature collapses into one halt-locus and species typing, and an open problem is exactly one missing bit, nothing more.

KEYWORDS *odd-supply separation; one-cut hypothesis; coordinate bridge; halt loci; species dichotomy; Millennium problems; barrier taxonomy*

1. BACKGROUND AND RATIONALE: THE BARRIER LANDSCAPE

Twentieth-century mathematics settled a question its nineteenth-century predecessor could not have posed: not whether particular problems are solved, but whether whole classes of methods can solve them. The answers form a scattered but growing inventory of impossibility theorems about technique families. Relativization bars every proof that is insensitive to oracle attachment from settling P versus NP (Baker–Gill–Solovay 1975). Natural proofs bar every constructive, large lower-bound property from separating circuit classes, conditional on pseudorandom generators (Razborov–Rudich 1997). Algebrization extends the bar to arguments invariant under algebraic extension (Aaronson–Wigderson 2008). In sieve theory, the parity problem bars classical sieve weights from distinguishing integers with even from odd numbers of prime factors (Selberg, mid-century), and the GPY optimality theorem proves that single-variable sieve weights below the $1/2 + \varepsilon$ level cannot deliver bounded prime gaps (Goldston–Pintz–Yıldırım 2005). In partial differential equations, an averaged analog of the Navier–Stokes equations exhibits finite-time blow-up while exhausting every known supercriticality barrier: energy-class, scaling, and symmetry arguments provably cannot settle global regularity by current routes (Tao 2016). In geometric complexity, occurrence-based vanishing arguments are refuted as routes to the flagship obstructions (Bürgisser–Ikenmeyer–Panova 2019). In Hodge theory, integral cycle constructions fail in general (Atiyah–Hirzebruch 1962; Kollár 1990). In smooth four-dimensional topology, the Whitney trick fails (Casson handles are merely topological), which is why the smooth Poincaré question in dimension four survives its

higher-dimensional siblings. In computability, generalized Collatz-type maps simulate arbitrary computation, with totality undecidable for the class (Conway 1972). In analytic number theory, Montgomery’s pair-correlation theorem holds only for test functions with Fourier support $|u| < 1$, the support boundary standing as a named gate (Montgomery 1973). And behind all of these sit the independence results: no proof within one consistent theory settles its own consistency (Gödel 1931), and combinatorial truths escape Peano arithmetic entirely (Paris–Harrington 1977; Kirby–Paris 1982).

Three observations convert this inventory from a list into a diagnosis problem. First, every one of these barriers is structural, not computational. None says that a technique is too slow or too weak at current parameters; each proves that a method class, idealized and closed under its own moves, is blind to the target in principle. Better computation moves constants, as the prime-gap bound moved from $p^{0.525}$ (Baker–Harman–Pintz 2001) through decoupling to $t^{13/84+\varepsilon}$ on the Lindelöf flank (Bourgain 2017), without touching the walls above. Second, the inventory has no common anatomy. Each barrier is proved in its own idiom, for its own field, against its own method class, and the literature carries no shared account of what a blocked corpus cannot express. Third, the failure mode, where one looks across fields, wears one shape: the barred methods are exactly those insensitive to a distinguished two-valued feature of the problem, its sign, parity, orientation, or truth direction. Oracle-invariant arguments cannot see the separation bit; parity-even sieve weights cannot see the prime versus P_2 flip; energy-class estimates cannot see the sign of the alignment defect; occurrence arguments cannot see multiplicities. The barrier is, in each case, an evenness statement about a reading class.

The standard model’s domain of validity can therefore be mapped precisely: within-class refinement succeeds exactly where the target is visible to the class’s own symmetry, and fails exactly where the target is odd under it. A barrier that will yield to better computation is one whose target the class can already read; a barrier requiring different geometry is one whose target the class is provably unable to read. What the literature lacks is not another barrier theorem but a theory of the bit the barriers are all blind to. That is the gap this paper addresses. The offering-bit program (Islam 2026a, author-self-citation, the framework under analysis throughout) supplies a finite, theorem-grade core for exactly this situation, and this paper tests that core’s central hypothesis against the dated corpus of twenty-three problem rows, with the grade discipline the program itself imposes: no instantiation on an open row is ever more than a structural diagnostic, and nothing here is a solution claim.

2. PRIOR APPROACHES

Six prevailing framings of blockage deserve their due before the synthesis.

The complexity barrier program (Baker–Gill–Solovay 1975; Razborov–Rudich 1997; Aaronson–Wigderson 2008) is the most explicit: it names method classes and proves their impotence. Its reliance, unexamined, is on the assumption that each barrier is a fact about computational proof technology specifically, rather than an instance of a domain-general phenomenon; the program never asks why the same evenness shape recurs in sieve theory, partial differential equations, and topology, fields with no proof complexity in common.

The sieve-theoretic response to the parity problem is the most successful bypass on record: multidimensional weights and smooth moduli escaped a proved optimality barrier (Zhang 2014; Maynard 2015), delivering bounded prime gaps. Its reliance is subtler: the bypass worked by changing the weight family, not by refining within the proved-inert one, and the theory internalizes this as technique rather than as law, so the parity problem persists at gap 2 even after the GPY wall fell (Maynard 2015).

The proof-theoretic independence line (Gödel 1931; Paris–Harrington 1977; Kirby–Paris 1982; Gödel 1938/40; Cohen 1963) locates obstruction at the axiom level. Its reliance is on a fixed formal system as the arena; it does not speak to method-class barriers inside fields whose obstacles are not provability-theoretic, and it offers no account of why some independences are later crossed by new axioms while method barriers fall to new objects.

The PDE criticality program, culminating in the averaged-equation blow-up theorem (Tao 2016), maps the exact reach of energy, scaling, and symmetry arguments. Its reliance is the implicit premise that the obstruction is a supercriticality exponent, a parameter, rather than a missing orientation: the theorem itself shows every argument in the barred class is even under critical scaling, unable to read the sign of the defect it averages away.

Geometric obstruction theory (Bürgisser–Ikenmeyer–Panova 2019; Atiyah–Hirzebruch 1962; Kollár 1990) proves named families non-supplying. Its reliance is the treatment of each refutation as terminal, a wall to route around, rather than as a datum about what the corpus contains: a proved non-supply is itself a dated entry about the register’s evenness.

The effectivity program around Hilbert’s sixteenth problem (Écalle; Ilyashenko, circa 1991 to 1992; Roussarie’s finiteness program) proves individual finiteness and pursues uniform bounds. Its reliance is the hope that quantifier order is a technicality: the entire open content is the swap $\forall\exists \rightarrow \exists\forall$, a two-valued feature, finiteness per instance against a uniform bound, that no amount of per-instance proof supplies.

Every prevailing approach shares one structural error, and it is the one Section 1 isolated: each treats its barrier as a fact about its own technique family and its own field, and none treats the barrier as an instance of a single shape, an even register blind to an odd target, missing exactly one binary coordinate. Section 4 develops that shape as mathematics; Sections 4.5 through 4.8 then show that the shape, once posited, absorbs each of these approaches as a case.

3. METHOD

The instrument is the odd-supply frame (Islam 2026a, Part One), used here exactly at the grades its author assigns it. Three methodological commitments govern every claim that follows, and they are stated as reproducible conditions.

The frame. A frame is a tuple $(X, \Gamma, C, d, \Delta)$: a finite data set X ; a finite group Γ acting on X ; a reading class C , the admissible corpus readings; a Boolean target $d : X \rightarrow \{0, 1\}$; and a dated corpus Δ , the record of what has entered and when. A reading is even under the row's native involution τ when it is constant on τ -orbits; the target is odd when it separates some orbit. The frame's finite theorems are stated in Section 4.1.

Grade discipline. Claims carry exactly three grades in this paper. Theorem-grade: proved in the finite core of the framework under analysis, and quoted as such. Hypothesis-grade: the One-Cut Hypothesis, never asserted as fact. Structural diagnostic: every instantiation on an open problem, without exception. The enforcing instrument is the coordinate bridge (Islam 2026a, Def. 3.2): an object domain enters the finite theory only through an equivariant map (B, φ) with factorization $d_M = d \circ B$ and an image clause, and a continuous parameter or Lie action is not a finite-group generator by declaration. Absent a bridge, no finite theorem is invoked on a row, and this paper supplies no bridges.

Three conditions for any row claim. Any claim about a problem row must satisfy: (1) formal derivation, the claim follows from the frame's finite theorems or is explicitly labeled structural; (2) corpus signature, the claim is checkable against the dated historical record of the row, entries, barriers, and crossings with dates; (3) invariance under reparametrization of the reading class, so that a renaming of methods cannot change the verdict, which is the operational content of the provenance clause. The falsifiable predictions of Section 5 are, per the framework's own architecture, testable independently: the BSD machinery is machine-checked (Islam 2026a, Appendix B, Lean and Fortran codebook), and the corpus audits of Section 4.6 are re-executable by any group against the published literature, two modalities addressing different validity tasks, machine checking the formal claims and corpus audit the application claims; their independence is structural and asserted, not demonstrated, and the corpus audits' decisive witnesses remain unexecuted, an aperture named in Section 6.

Modeling freedom, stated. The frame's components and the per-row choices of register, target, and admissible supply are structural degrees of freedom, not fitted numerical constants: each is fixed and recorded per row before typing, in the target and register columns of Tables 2 and 3, and their effective freedom is not quantified. The twenty-three rows are retrospective classifications against the dated corpus, not twenty-three independent predictions; the paper's prospective content is exactly the two predictions of Section 5. Each row is typed under its single declared encoding as printed; invariance of a row's species and bit under alternative admissible encodings is unproved, the admissibility-equivalence constraint is owed, and a row exhibited with two natural encodings yielding different species or different bits loses its typing.

What this method does not do. It proves nothing about zeta zeros, vorticity, Hodge classes, ranks, mass gaps, complexity classes, or Ricci flow. It assigns each open row a position in a typed obstruction space and names, for each, the single object whose dated entry would constitute a crossing. The distinction between assigning a position and solving a problem is the paper's load-bearing fence, and it is enforced by the falsifiers of Section 4.3, which are armed against the framework itself.

4. THE CORE: ONE BIT ACROSS TWENTY-THREE ROWS

4.1 The finite core

Four theorem-grade results of the framework carry the entire paper (Islam 2026a, Part One, quoted at their stated grades).

Block theorem (Thm. 4.1). On a finite frame, under the native involution τ , the reading space splits as $V = V_+ \oplus V_-$ into even and odd parts; even readings factor through the orbit space, and an odd target is decided by no even reading. Even-register readouts are blind to odd targets: this is the parity law, and every barrier in Section 4.7 will be read as an instance of it. Two senses of odd are kept fixed throughout: a Boolean target is odd when it separates some orbit (non-invariance, $d \circ \tau \neq d$), and a sign vector is wholly odd when $t \circ \tau = -t$; the block theorem needs only the former, a nonzero odd component already barring decision by even readings.

Crossing theorem (Thm. 4.2). A target odd under τ is decided exactly when a supply enters that is odd under τ and independent per orbit; the crossing is an entry, not a refinement. The decoder is stated, not assumed: at the seat the supply separates the obstructing pair, $s(x) \neq s(\tau x)$, and the target is recovered as the supply's sign composed with one calibration, the dated entry record's identification of which sign is which. Oddness of a supply that does not separate the target's seat decides nothing, and antisymmetry of supply and target on disjoint orbit sets is not a crossing. Independent per orbit is the framework's term (Islam 2026a, Part One), pinned here: the even readings, constant on orbits, carry no coordinate of the odd part at any orbit, so the supply's odd content at each orbit is information the register does not contain. Necessity is the parity law; the sufficiency direction is quoted at the companion's theorem grade with the decoder as just stated, no claim being made for arbitrary decoders or for supplies separating orbits the target does not separate.

Representation supply (Thm. 4.3) and the one-bit corollary (Cor. 4.4). On each free orbit the odd part is one-dimensional, so the missing coordinate at that orbit is exactly one bit: $\dim V_- = 1$ per free orbit, and the decision reduces to supplying the sign. The unit of the corollary is the row's single target, and the deficit is measured at the target's seat, the orbit the target separates: at that seat the missing coordinate is one binary orientation, never two, never a continuum. A frame with several free orbits carries an odd function space of one dimension per orbit; the corollary sizes the seat of one Boolean target, not the whole space. The per-row claim carries one stated typing premise, the seat-uniqueness condition: a row is typed one-bit when its named Boolean target, conditioned on the dated corpus, has exactly one unresolved orbit orientation, equivalently when the corpus-compatible target set is exactly $\{t_0, -t_0\}$ for a reference target t_0 fixed by the row's prior information. With m independent unresolved orbit orientations the compatible set has 2^m elements and the row would carry m bits, not one. The premise is checked per row in the target columns of Tables 2 and 3, where each row's target is one named Boolean decision, and it is armed: a row whose compatible target set is exhibited to exceed two falls out of the exact typing to one bit per unresolved orbit, and the universal invariant of Section 4.8 falls with the first row where no such seat exists. The bit prices the decision's output at the seat, the sign; it does not price the construction cost of the supply, which is unbounded and row-specific. Where a row's missing supply is not Boolean the row is a wall row under Section 4.4, the bit then the orientation of a constructed object rather than the sign of a real coefficient.

Provenance collapse (Thm. 4.5). Inside the fixed grammar G_0 , the deletion test is decidable under the companion's stated effectiveness premises, a finite effectively enumerable reading class with terminating evaluations (Islam 2026a, quoted at theorem grade with those premises); outside them decidability is not asserted here. A candidate supply whose deletion from the corpus leaves every reading unchanged is a restatement, not a supply. This is the contamination barrier; it is what keeps a corpus honest about what has actually entered.

Two further results carry the dynamics. **First-sufficient-entry (Thm. 5.1)** orders crossings by sufficiency. **Seat-hiding (Thm. 5.3)** proves that no halting router without a seat bit, the router's own record of whether a crossed witness has entered, can distinguish histories differing only in a crossing: a router returning the same token on histories that differ only in whether the witness entered is provably not tracking the corpus state. Theorem 5.3 is the load-bearing certification instrument of Section 4.6.

4.2 The dated-entry recursion and the Ninth Gate

The framework's third component turns static separation into a dynamical account (Islam 2026a, Part Three, quoted at its grades). The dated-entry operator T maps a corpus to its successor whenever a supply enters: odd under τ , independent per orbit, provenance-clean (Def. 4.1). Strict refinement (Part Three, Lem. 4.1) and halting within $|X|$ entries on a finite frame (Part Three, Thm. 4.2) are theorem-grade, as is passivity: only the corpus lattice moves under an entry; the register's laws never do (Part Three, Cor. 4.3). The halting typing (Part Three, Thm. 4.4) classes every odd target at the halt state as exactly one of three: **crossed**; **restatement-blocked**, every candidate supply already corpus-contained, so the deletion test fires; or **defeater-absent**, identified with the diagonal species of Section 4.3. Without the provenance clause every finite free frame crosses via a transversal selector, so all content lives in provenance against a real dated corpus.

The entry itself is typed by the Ninth Gate port (Islam 2026a, Section 5, structural grade): one diagonal, two surfaces. The wall surface: no index carries the diagonal of negation, and a record of a deed across a domain the register cannot absorb meets the wall and is not an object of the reading register. The self surface: negation's only fixed point is the unsettled value. The port is not a metaphor appended to the theory; it is the theory's typing of where a deed can enter from, and it is the structure Sections 4.4 and 4.8 read off the continuous and undecidable rows.

4.3 The One-Cut Hypothesis and its armed gates

The framework's central posit (Islam 2026a, Hyp. 11.1, structural form; Hyp. 11.2, grand form) states: every blocked row is an even register blind to one odd target, and the row resolves by the dated entry of one provenance-clean supply carrying the missing orientation coordinate; the cut severs the decision bit and does not replace object-level construction (Section 11.5). The hypothesis stands at hypothesis grade throughout this paper.

Its falsifiers are not vulnerabilities but armed enforcement gates, each naming the exact observation that would break one named claim and nothing above it (Islam 2026a, F1 to F6):

Table 1 | *The six enforcement gates of the One-Cut Hypothesis, each with its named blast radius. (tier: hypothesis with named falsifier)*

Gate	Firing observation	What falls
F1	An even corpus reading decides an odd target	The parity law's application; everything collapses
F2	A proved-inert named family supplies after all	The restatement-blocked typing of the row
F3	A method-blocked row has no pair defeater	The census leg; the row falls to the diagonal species
F4	A row's missing supply is provably non-orientation, purely continuous with no Boolean decision seat	The literal one-bit reading of that row; the hypothesis holds in the after-construction reading
F5	A constructor outputs a restatement	The grand form
F6	A crossing enters without provenance	The dated-entry account of that row

Two consequences bind the rest of the paper. The species dichotomy (Islam 2026a, Section 15.1, an external hypothesis-grade input) classes every resistance as common, admitting a pair defeater, or diagonal, defeater-absent; every species typing below inherits that input's grade. And the gates are what make the mapping falsifiable: each row assignment of Sections 4.5 through 4.7 carries its nearest gate, so each is an exposed, checkable structural claim rather than a relabeling.

4.4 The wall: continuous domains and the orientation law

The coordinate bridge (Def. 3.2) is the exact boundary where discrete reading registers meet continuous object domains. The impossibility of absorbing a real-coded continuum into a finite register without an equivariant bridge is not a limitation of the mapping; it is the one-bit separation itself, stated at structural grade. In the dynamical account this is the wall surface of the Ninth Gate: a record of a deed across a continuous domain meets the wall and is not an object of the reading register (Islam 2026a, Section 5).

The governing law for such rows is **orientation-after-construction, never orientation-instead-of-construction** (Islam 2026a, Section 11.5). Where a row's missing supply is not Boolean, object-level construction remains: Navier–Stokes its data-uniform modulus, Hodge its semiregular witness, Yang–Mills its scale-invariant monotone functional, the Riemann Hypothesis its operator. Falsifier F4 is the armed enforcement gate of this law: the necessity of object-level continuous construction does not dilute the model; it fixes the one-bit decision seat strictly at the post-construction boundary. The literal and after-construction readings are two claims with two survival assessments, stated before any firing and never adjusted after one: on the wall rows the after-construction reading is the primary typing, not a retreat, and the literal reading is asserted nowhere on them. On the wall rows the parity law is read in diagnostic form, no limit system and no preservation theorem supplied or claimed: the diagnostic content is that no internal refinement of a continuous method class substitutes for the missing provenance-clean supply. Nothing escapes. The bit is one, the wall is the bridge gate, and what is missing on these rows is not vocabulary but the deed.

4.5 The twenty-three rows

Tables 2 and 3 assign every row its gate typing. The typology is one-bit throughout: a row is **exact** where the corpus holds a Boolean decision bit and a named, dated even register; a **wall row** where the carrier is continuous, the one-bit seat fixed after construction by the F4 gate; a **pure unbridged target** where the corpus holds no even register at all, the separation law's domain boundary: with no specified register carrying the relevant invariance the law has no instance on the row, applicability is unresolved by construction, and the typing records exactly that absence, a block by missing register rather than by evenness; a **diagonal-port row** where the habitat is

defeater-absent under F3; and **one bit from closure** where the flip or boundary is named, the register partial, and the carrier absent. These are positions on the separation law’s boundary, not grades of framework success.

Table 2 | *The seven Millennium rows under the frame.* (tier: structural diagnostic; the Poincaré row is a crossed control)

Row	Target bit	Native flip / even register	Owed supply	Gate typing
P versus NP	separation sign	three barrier classes even under their flips	obstruction family with orientation	exact (structural)
Riemann Hypothesis	the global decision, zeros on the line	Davenport–Heilbronn register (1936)	the operator; orientation-after-construction	exact (structural)
Navier–Stokes	breakdown sign	energy and harmonic-analysis arguments even under critical scaling (Tao 2016)	data-uniform modulus dominating the alignment defect	wall row (F4 gate)
Yang–Mills	mass-gap sign	strong-coupling alternation; continuum-limit arguments	scale-invariant monotone supply, infrared end trivial and gapped	wall row (F4 gate)
Hodge	class sign	integral constructions fail (Atiyah–Hirzebruch 1962; Kollár 1990)	semiregular witness	wall row (F4 gate)
BSD	rank parity	root number, parity of analytic order	height pairing nondegenerate at order two, built	exact at orders 0, 1 (beachhead); owed at order 2
Poincaré (crossed control)	noncollapsing	parabolic-rescaling reading class	provenance-clean W-functional and reduced length, entered 2002 (Perelman)	exact (crossed; seat-bit certified)

Each row’s target is fixed as printed in the target column and nowhere shifts: the Riemann row’s single Boolean is the global decision, all nontrivial zeros on the critical line, the per-zero truth directions its indexed decomposition, one local decision never substituting for the conjunction.

The BSD row is the framework’s beachhead: the unique open row whose corpus already records two dated entries of the supply shape (Islam 2026a, Part Three, Section 7). Gross–Zagier (1986) equated the Néron–Tate height of a Heegner point with $L'(E, 1)$, the first bridge from analytic rank-one data to an actual rational point, and Kolyvagin (1990) supplied the Euler system converting the construction into the rank identity with Sha finite at analytic order at most one. The paper types both as supply-shaped, the Heegner construction living exactly on the odd side of the row’s native flip. The order-two barrier is named exactly: the Heegner construction is inert above rank one, and the owed offering is a height pairing nondegenerate at analytic order two, built by a construction fixed in advance from data consulting neither rank nor Selmer group, the route-axiom Ω . The row’s typed target is the parity bit alone: rank identity and finiteness of Sha are separate predicates of the companion’s conditional machinery, not content of this row’s bit, a parity bit not determining a rank identity, ranks zero and two sharing parity. The conditional machinery is machine-checked (Islam 2026a, Appendix B.5): Ω with the inputs yields the object; bare Ω is the object in disguise; no reading of the inputs alone decides the object, witnessed by a model swap; and Ω types as an odd supply on the model. Executed on the row’s own data, the conjecture returns WORLD-ROWED on the rank identity and finiteness at analytic order at most one, and UNPOPULATED at order two and above uniformly, the exhibition row a deed populates curve by curve and only a construction populates uniformly.

Table 3 | Sixteen extension rows under the frame, with the logical type of each statement (Π_1 finitely refutable by deed; Π_2 possibly true-but-unprovable; analytic, quantifying over a real-coded continuum). Flag (heuristic) inherits its source's verification status. (tier: structural diagnostic)

Row	Logical type	Native flip / dichotomy	Register status	Owed supply	Gate typing
Goldbach	Π_1	prime versus P_2	parity problem named (Selberg)	parity-breaking distribution input	exact (structural)
Twin primes	Π_2	parity gap-2 parity	Maynard–Tao inertness proved at gap 2	distribution input beyond the proved-inert family	exact (structural)
Legendre	Π_1	prime in $[n^2, (n+1)^2]$ per instance	conditional supplies stop short (Cramér 1936; Baker–Harman–Pintz 2001)	sub-square-root distribution input	exact (structural)
abc	Π_1 per instance	quality $q > 1$ flip	Stewart–Yu plateau named	radical-quality supply above the plateau	exact (structural)
Jacobian	Π_1 per instance	invertibility sign	reduction machinery named	degree-2 certificate or obstruction	exact (structural)
Mersenne primes	Π_2	both polar answers open	none dated on either side	any supply into sparse exponential sequences	pure unbridged target
Odd perfect numbers	Π_1 (nonexistence)	deficient versus abundant sign	abundancy fixed point $I(n) = 2$; floor $N > 10^{1500}$ (Ochem–Rao 2012)	none established	pure unbridged target
Smooth 4D Poincaré	effectively Π_1 -like (heuristic)	exotic S^4 existence versus nonexistence	Whitney trick fails in dimension 4	finite handle-diagram deed or obstruction theory	pure unbridged target
Collatz	Π_2	cycle versus divergence, not an involution	Terras 1976 and Tao 2019 ceilings proved	cycle-exclusion deed or divergence witness	diagonal-port row (F3)
Hadwiger	Π_1	minor/complement duality (Steiner 2021)	live register, dated entries 2019 to 2021	linear-bound deed	one bit from closure
Sunflower	Σ_2	C^k versus $(\log k)^k$ bounds	robust-sunflower machinery partial (Alweiss–Lovett–Wu–Zhang 2021)	the constant $C(r)$ itself	one bit from closure
Erdős–Straus	Π_1	exception set empty versus density zero (Vaughan 1970)	modular-identity covering class partial	identity family covering all residue classes	one bit from closure
Beal	Π_1	hyperbolic flip boundary $\sum 1/x < 1$ (Darmon–Granville 1995)	Frey-curve class named	uniform Frey-type parametrization	one bit from closure
Invariant subspace (Hilbert)	analytic	Banach flank negative (Enflo 1976/87; Read 1984), Hilbert flank open	Lomonosov-type register named	cyclic-vector operator or new positivity theorem	wall row: the bridge gate is the bit
Hilbert's 16th, second part	Σ_2 over reals per n	individual finiteness versus uniform bound	non-accumulation proofs, non-effective	effective cyclicity bound	wall row: the bridge gate is the bit
Lindelöf / Montgomery PCC	analytic	zeros-to-primes duality (Goldston–Montgomery 1987)	moment and decoupling register partial (Bourgain 2017); support $ u < 1$	support-boundary crossing supply	wall row: the bridge gate is the bit

The wall rows of Table 3 are where the coordinate bridge bites hardest, and the bite is the claim. Each quantifies over a real-coded continuum, and a continuous parameter is not a finite-group generator by declaration; hence no finite frame candidate exists without a bridge this paper does not supply. The fields themselves have internalized the flip structure: Montgomery's pair-correlation theorem holds for test functions with Fourier support $|u| < 1$ under RH, the support boundary being the field's own named gate, and the Goldston–Montgomery equivalence (1987) between pair correlation and the variance of primes in short intervals is a genuine two-way duality,

object-level mathematics wearing the flip shape. Hilbert’s sixteenth isolates the purest quantifier obstruction: individual finiteness of limit cycles is proved, the uniform bound $H(n)$ is open even at $n = 2$, and the entire open content is the swap $\forall\exists \rightarrow \exists\forall$. The invariant subspace row exhibits answer instability under a small domain flip, false for general Banach spaces and open for Hilbert space, with both flanks’ method classes named and neither transferable. On all three rows the typing is the framework’s central structural claim located exactly where the hypothesis says the wall must stand.

4.6 Crossed controls: seat-bit certifications

The framework claims a resolution enters a blocked corpus in a fixed shape: the dated entry of one decisive object, odd under the row’s native flip, provenance-clean, whose certificate the prior reading class could not have denoted (Islam 2026a, Def. 6.1). The shape is testable where testing is cheap, on rows already crossed. Nine historical events, ten dated entries (the BSD event contributing two, the Gross–Zagier and Kolyvagin rows of Table 4), were checked against the checklist: blocked corpus; non-denotable decisive object; dated entry; provenance-clean; later corpus verification.

Table 4 | *Ten dated crossings as seat-bit certifications: in each, the corpus’s routing flipped only at a dated, provenance-clean entry whose decisive object the dated prior register is not recorded to have denoted (Thm. 5.3 the conditional instrument). (tier: documentary certification with the Cor. 5.2 caveat)*

Crossing	Blocked corpus	Decisive object	Entry dated	Why the prior register was blind
Perelman, Poincaré and geometrization	Ricci flow could not control collapsing	$\mathcal{W}$ -entropy, reduced length, reduced volume	11 November 2002; March, July 2003	Pre-2002 reading class even under parabolic rescaling; sources absent from the flow literature
Gross–Zagier, BSD rank one	no object encoded L' -data as a point height	Heegner-point height formula	1986	Analytic register carried no height pairing
Kolyvagin, BSD rank identity	Selmer bounds uncontrolled	Euler system of norm-compatible points	1988/1990	No norm-compatible system in corpus
Wiles and Taylor–Wiles, Fermat	no modularity route	deformation rings; the 3-5 switch	1994/1995	Switch not recorded in the prior deformation class; non-denotability conditional (Thm. 5.3)
Green–Tao, primes in progressions	Szemerédi only at positive density	transference principle; GPY pseudorandom majorant	April 2004	No majorant met the pseudorandomness line
Zhang to Maynard, bounded gaps	GPY optimality proved for single-variable weights	multidimensional weights; smooth moduli	2013/2014	The inert family was a theorem, not a practice
Hales–Ferguson, Kepler	score bounds hand-checked	purpose-built score; Flyspeck certificate	1998/2014	Certificate not recorded until formalized; non-denotability conditional (Thm. 5.3)
Appel–Haken, four color	human survey of configurations	machine certificate	1976	Certificate beyond hand-survey readings on the dated record; non-denotability conditional (Thm. 5.3)
Tao, discrepancy	positive-density arguments	logarithmic averaging as missing weight	2015	Averaging register not in corpus
Gödel, incompleteness	Hilbert’s program	arithmetized self-referential sentence	1931	Diagonalization required a coding technology invented inside the proof

Every historical crossing certifies the same entry shape at documentary grade: a dated, provenance-clean entry, the routing flip at the entry and not before. The non-denotability leg is conditional content under Thm. 5.3, whose frozen-grammar witnesses are unexecuted, and that aperture is named in Section 6. Documentary absence from the dated record is not non-denotability: this paper claims the first and prices the second as conditional. The framework’s own caveat rides with the certification and is part of its honesty: retro-typing mints no new object-mathematics and attributes no intent (Islam 2026a, Cor. 5.2), and a shape fitting every crossed row can fit zero open rows. The controls are certification of the entry law, not inductive support for the One-Cut Hypothesis, which remains hypothesis-grade.

4.7 The barrier taxonomy mapped

Seventeen external barriers type under the halt-locus and species vocabulary with no remainder except theory-side independence, which occupies the diagonal habitat.

Table 5 | *The barrier literature under the frame: method class ruled out, paper-side typing, nearest enforcement gate, and named crossing or escape.* (tier: structural diagnostic; crossing column documentary)

External barrier	Method class ruled out	Frame typing	Gate	Named crossing or escape
Relativization (Baker–Gill–Solovay 1975)	diagonalization, simulation	reader-barred; common; oracle pair as defeater	F1	IP = PSPACE (Shamir 1990)
Natural proofs (Razborov–Rudich 1997)	constructive large properties	reader-barred; common; deletion-test mirror	F5	Williams ACC bounds (2011)
Algebrization (Aaronson–Wigderson 2008)	all algebrizing arguments	reader-barred; common	F1	Williams code access (2011)
Incompleteness (Gödel 1931)	all proofs within one consistent theory	terrain-empty in-system; diagonal habitat	F3	none internal; new axioms only
Paris–Harrington (1977); Goodstein (Kirby–Paris 1982)	PA-provable Ramsey and termination	theory-side wall; diagonal habitat	F3	provability in stronger systems
Continuum hypothesis (Gödel 1938/40; Cohen 1963)	all of ZFC	terrain-empty for ZFC; diagonal	F3	none
Resolution lower bound (Haken 1985)	polynomial-size Resolution for PHP	reader-barred over Resolution; common	F1	Frege program, open
Bounded-arithmetic unprovability (Ajtai 1994)	constant-depth Frege	reader-barred; theory-side	F3	exponential sharpenings
Naturalized-proofs mirror (Razborov 1995)	bounded-arithmetic lower-bound proofs	deletion-test mirror, theory-side	F5	width machinery; Frege open
Sieve parity problem (Selberg)	classical sieve weights, prime/ P_2 flip	witness-absent; common; parity-even register	F1	none at gap 2; bounded gaps bypass (Maynard 2015)
GPY optimality (2005)	single-variable weights below $1/2 + \varepsilon$	restatement-adjacent; proved non-supply of named family	F2	multidimensional weights (Maynard 2015); smooth moduli (Zhang 2014)
Averaged-NS blow-up (Tao 2016)	energy, scaling, symmetry arguments in 3D	witness-absent, inert; energy register even under critical scaling	F4	none; algebraic cancellation required
Occurrence refutation (Bürgisser–Ikenmeyer–Panova 2019)	occurrence arguments	proved non-supply of occurrence family; restatement-adjacent	F2	multiplicity obstructions
Integral Hodge failures (Atiyah–Hirzebruch 1962; Kollár 1990)	integral cycle construction	witness-absent; common; semiregular-deformation register even	F4	rational conjecture survives
Whitney-trick failure, dimension 4	smooth surgery and h-cobordism	reader-barred over smooth-surgery class; witness-absent	F3	topological category only (Freedman 1982)
Conway universality (1972)	uniform decision methods for Collatz-type maps	diagonal habitat, meta-level; theory-side	F3	none known
Montgomery support boundary (1973)	pair correlation beyond tested support	wall row: the bridge gate is the one-bit separation itself	F4	none named

The gate column separates the barrier species cleanly: method barriers sit nearest F1 or F5, theory-side walls nearest F3, continuous-supply rows nearest F4, exactly the distribution the framework’s own order of work predicts, with F4 examinations first for Navier–Stokes and Hodge. Every named crossing is a deed-shaped escape entering the corpus dated and outside the barred class, never obtainable by tuning the proved-inert family; hence the two barriers whose inertness is itself a theorem, GPY optimality and the occurrence refutation, are the strongest analogues of the restatement-blocked mode. The empty crossing cells are as informative as the full ones: no independence result has been crossed from inside its theory, and no named escape exists for the parity problem at gap 2 or the averaged-equation barrier.

4.8 The universal one-bit invariant

Three axes of difference organize the twenty-three rows, and none of them widens the cut. The supply-type axis separates Boolean decision bits from continuous carriers; the logical-type axis separates finitely refutable Π_1 statements from Π_2 statements that could be true-but-unprovable and from analytic statements over real-coded continua; the flow axis separates rows adjacent to a scale-invariant monotone supply from rows with no named flow. Along every axis the structural deficit stays exactly one bit wide at the row's decision seat, the seat-uniqueness premise of Section 4.1 carried by every row typed. What changes across the axes is not the width of the cut but the register's illusion of self-sufficiency: intact for Π_1 rows with named registers, shaken for Π_2 rows in the diagonal habitat, and gone entirely at the wall, where the analytic rows stand. This is the comparison's substantive finding. The isomorphism claim is exact at the level of block and sufficiency shape for every row possessing a Boolean decision bit and a named, dated even register, six of seven Millennium rows structurally and the strong sieve family among the extension rows; and at the level of carriers the one-bit invariant holds without exception, read always at the decision seat under the same premise, the wall rows, unbridged targets, and diagonal-port rows differing in which gate they stand at, never in the width of what is missing.

5. FALSIFIABLE PREDICTIONS

Two predictions, each forced by the framework's own geometry, each seed-native, each conditioned, each aimed at a named claim of this paper and nothing above it. Both run against the author and are reported without softening. Forced has an exact scope in this section: the prediction follows from the row's typing under the framework's stated laws, conditional on that typing; no finite theorem is invoked on any unbridged row (Section 3), and nothing here is a theorem about an object domain. Each prediction stands at structural-diagnostic grade with its kill-switch named.

P1, the order-two beachhead (forced by the BSD typing). Prediction: for elliptic curves of analytic order two, a height pairing nondegenerate at order two exists and is constructible by a procedure fixed in advance from data consulting neither the rank nor the Selmer group, the route-axiom Ω of Section 4.5. Typing: structural-necessity prediction forced by the beachhead reading, the forcing internal to the framework and conditional on the typing; seed-native, resting on no imported premise beyond the dated Gross–Zagier and Kolyvagin entries. Binding conditions: the prediction quantifies over analytic order exactly two and binds at order two, where the Heegner construction is proved inert; it says nothing at orders zero and one, where the row is already crossed, and asserts nothing at orders three and above, where no typing is claimed. Confirmation threshold: an exhibited construction, executed on a named curve of analytic order two, returning a nondegenerate pairing; the confirmation is curve-by-curve exhibition plus a uniform construction. Kill-switch: an exhibited obstruction to a nondegenerate order-two pairing resolves the beachhead downward (Islam 2026a, Part Three F3, the beachhead falsifier), and with it falls this paper's claim that BSD is the framework's designated test bed. The danger is named: the owed construction may not exist, and the framework's own machinery is what would certify its absence. No necessity theorem is claimed or held: P1 asserts that one construction of the named shape exists and is constructible under the route-axiom conditions, not that every resolving supply factors through it; the conditional machine-checked result, Ω with the inputs yields the object, does not establish Ω , and Section 4.5 says so. The failure of one algorithm fires nothing; only an exhibited obstruction to a nondegenerate order-two pairing fires the kill-switch.

P2, the next crossing enters non-denotable (forced by Thm. 5.3 and the entry law). Prediction: the next crossed row among the twenty-three of Tables 2 and 3 will enter through a dated, provenance-clean object whose certificate the pre-crossing even register could not denote, and the corpus's routing will flip at that entry and not before. Typing: structural-necessity prediction forced by the seat-bit theorem and the crossing theorem jointly, the forcing internal to the framework and conditional on the typing; seed-native, carrying no import. Binding conditions: the prediction binds on crossings, not on attempts, and a row that never crosses neither confirms nor fires it. Confirmation threshold: a retro-type audit of the crossing, of the kind Table 4 executes ten times, showing the decisive object non-denotable in the prior reading class and the routing flip dated to the entry. Kill-switch: a crossing whose decisive object was already denotable within the prior even register, an F1-type event, breaks the census leg of the framework, and with it falls this paper's claim that the controls certify the entry law. One dependence is named at its honest weight: P2's sharpness rests on an executable separator this paper does not supply, a formal denotability predicate relative to the prior register and a frozen witness family exhibiting prior-image exclusion. Until those are exhibited, P2 stands as a conditional template with a named confirmation protocol and a named kill-switch, and no necessity over future crossings is claimed: the seat-bit and crossing theorems force the entry shape inside the framework's typing and nothing about the object domain. Beneath the register, one tripwire at its honest grade: the species dichotomy is an external hypothesis-grade input

(Islam 2026a, Section 15.1), and its failure would re-type every diagonal-port row without touching the finite core.

6. DISCUSSION

The consequences cascade in three directions. First, for the barrier literature: the seventeen barriers of Table 5 cease to be seventeen facts and become one typing, and the practical question on any blocked row becomes which gate it stands at, since the gate dictates the shape of a crossing. A row at F1 wants a non-relativizing object; a row at F2 wants a supply outside the proved-inert family; a row at F3 wants a defeater or a new axiom; a row at F4 wants a construction, after which the orientation bit can exist. Second, for the open rows themselves: the mapping names, for each, the single object whose dated entry would constitute a crossing, from a parity-breaking distribution input for Goldbach to an effective cyclicity bound for Hilbert's sixteenth to the order-two pairing for BSD. The framework's value to a working mathematician on any of these rows is a specification of what to build, stated sharply enough to fail. Third, for the analytic rows: a research program is named that the literature does not carry, bridge engineering, the explicit construction of equivariant coordinate bridges from continuous object domains to finite frames, with the image clause doing the work. Whether any analytic row admits such a bridge is open, and Section 7 returns to it.

Anticipated objections, engaged on merit. That the controls retro-type and do not predict is true and is the framework's own Cor. 5.2, stated in Section 4.6; the certification claim is exactly what Thm. 5.3 entitles and no more. That the species dichotomy is an external input is true; every species typing inherits its grade, and the tripwire is named in Section 5. That the deletion test's flagship candidates, Miller's GRH-conditional primality reduction and Lagarias's RH criterion, remain unexecuted inside the fixed grammar G_0 is true; the restatement-blocked typings carry that aperture openly. That the framework might merely re-describe the obstruction landscape is the live question, and it is precisely what the two predictions of Section 5 exist to settle: a vocabulary that only re-describes cannot be fired by an exhibited obstruction or a denotable crossing, and this one can.

Limitations are named as grade boundaries, not breaches. No finite theorem of the framework was invoked on any open row, the coordinate bridge barring the propagation by declaration. The One-Cut Hypothesis is hypothesis-grade throughout. Every row assignment is a structural diagnostic. Nothing in this paper is a solution claim on any of the twenty-three rows, and the framework's enforcement gates are armed against the framework itself before anyone else. One validation this paper does not execute is named as an open aperture: a frozen-register benchmark on the restricted GPY method family, its admissible operations frozen from pre-crossing dated sources, its limitation formalized, and independent reviewers classifying the Maynard crossing without register redefinition; until executed, application validity rests on the documentary record of Section 4.6 alone. A second owed object is the independence decomposition of Section 3's two modalities: an explicit split of assumptions into shared and channel-specific parts, each channel's acceptance functional independent of the other channel's assumptions, is the owed demonstration, and until exhibited the modalities' independence remains asserted, as Section 3 prices it.

How the paper stands to each prior position. A statement is owed because this paper sits across several literatures at once, complexity barriers, sieve theory, proof theory, partial differential equations, arithmetic geometry, and topology, and its contribution read from outside is exactly the set of relations it carries to prior positions and nothing wider. Eight relation words, defined once and used as defined in every row of Table 6. **Additive:** the paper supplies a result or a test the position lacked and leaves it standing. **Replacing:** a framing is retired and a measured object put in its place. **Subsuming:** the prior claim becomes a case of the paper's object; in this paper every subsuming row is argued, never executed, the instance-map being Table 5's typing in words with no formal mapping supplied, so the word means candidate subsumption throughout and demotes en bloc if a mapping fails. **Corroborating:** independent agreement, which confers no warrant on either side and is counted as none. **Contradicting:** a named thesis denied on executed data, never on argument alone. **Competing:** a different position on the same question, argued and not executed. **Scoping:** the prior claim kept inside a stated boundary. **Kin:** a position the paper is continuous with, cited as such. The ninth word, superseding, is used nowhere in this paper: the paper retires no theory's central claim on executed data, and it supersedes nothing. The census: nine positions subsumed (candidate, as just defined), three scoped, six corroborated, one kin; none replaced, none contradicted, none competing, none additive. The paper's contribution is exactly the set of relations the table states, and nothing wider.

Table 6 | *The standing table: one row per prior position the body engages, relation words as defined above, evidence words from the closed list (documentary, cited, argued); no executed battery is claimed by this paper. (tier: structural diagnostic)*

Prior position	What it holds	What this paper does with it	Relation	Evidence
Relativization (Baker–Gill–Solovay 1975)	Oracle-invariant proofs cannot settle P versus NP	Types it as a reader-barred halt locus with the oracle pair as defeater, gate F1	subsuming	argued
Natural proofs (Razborov–Rudich 1997)	Constructive large properties self-defeat	Types it as the deletion test’s mirror on the theory side, gate F5	subsuming	argued
Algebrization (Aaronson–Wigderson 2008)	Algebrizing arguments cannot separate	Types it reader-barred over the algebrizing class, gate F1	subsuming	argued
Sieve parity problem (Selberg)	Classical weights are parity-even	Reads it as the canonical even register under the prime/ P_2 flip	subsuming	argued
GPY optimality (2005)	Single-variable weights provably stop short	Reads it as a proved non-supply, restatement-adjacent, gate F2	subsuming	argued
Bounded gaps (Zhang 2014; Maynard 2015)	New weight families escaped the wall	Certifies the escape as deed-shaped, outside the inert family	corroborating	documentary
Averaged blow-up (Tao 2016)	Energy-class arguments are exhausted	Types it witness-absent and inert, gate F4, algebraic cancellation owed	subsuming	argued
Occurrence refutation (Bürgisser–Ikenmeyer–Panova 2019)	Occurrence arguments provably fail	Types it as proved non-supply of the occurrence family, gate F2	subsuming	argued
Integral Hodge failures (Atiyah–Hirzebruch 1962; Kollár 1990)	Integral constructions fail in general	Types the row witness-absent with the semiregular-deformation register even, gate F4	subsuming	argued
Independence results (Gödel 1931; Paris–Harrington 1977; Cohen 1963)	No proof inside the system	Places them in the diagonal habitat, theory-side, gate F3, the paper’s one remainder	scoping	argued
Proof complexity bounds (Haken 1985; Ajtai 1994)	Resolution and bounded arithmetic are too weak	Types them reader-barred, theory-side, gates F1 and F3	subsuming	argued
Whitney-trick failure (Casson; Freedman 1982)	Smooth surgery fails in dimension 4	Types SPC4 as a pure unbridged target, reader-barred over the surgery class	scoping	argued
Conway universality (1972)	Collatz-type totality is undecidable in general	Places Collatz at the diagonal port, meta-level, gate F3	scoping	argued
Pair correlation (Montgomery 1973; Goldston–Montgomery 1987)	Support boundary $ u < 1$; zeros-to-primes duality	Names the boundary the field’s own wall; the row is a wall row, the bridge gate the bit	kin	cited
BSD entries (Gross–Zagier 1986; Kolyvagin 1990)	Rank-one identity with Sha finite	Certifies both as dated supply-shaped entries; the order-two pairing is owed	corroborating	documentary
Perelman (2002/03)	Geometrization via the W-functional	Certifies the crossing as the Grand Witness, seat-bit execution	corroborating	documentary
Modularity (Wiles 1995; Taylor–Wiles 1995)	Semistable modularity, hence Fermat	Certifies the entry shape, deformation rings and the 3-5 switch	corroborating	documentary
Primes in progressions (Green–Tao 2008)	Primes contain long progressions	Certifies the transference entry, provenance imported across literatures	corroborating	documentary

Prior position	What it holds	What this paper does with it	Relation	Evidence
Formal proof (Hales–Ferguson 2006; Hales et al. 2017; Appel–Haken 1977)	Machine-checked certificates settle	Certifies non-denotability of the certificate until formalized	corroborating	documentary

Every relation word above corresponds to a body sentence carrying that relation at that grade; every evidence word matches the tier the body carries for that position; no figure in any row departs from the body’s rounding. A row found inflated is repaired by demotion of the word, never by promotion of the body.

7. CONCLUSION

The gap this paper addresses is anatomical: mathematics possesses a growing inventory of proved method-class barriers and no shared account of what a blocked corpus cannot express. The proposed resolution is a single structure, the odd-supply frame with its parity law, its one-bit corollary, its dated-entry recursion, and its Ninth Gate port, under which every mapped barrier is an evenness statement and every open row is missing exactly one binary orientation coordinate. The mapping of twenty-three rows and seventeen barriers is executed at structural grade, with the framework’s finite theorems nowhere invoked on open rows and the One-Cut Hypothesis nowhere asserted as fact.

The primary falsifiable prediction is P1 of Section 5, forced within the framework’s typing and conditional on it, as Section 5 scopes the word: a provenance-clean height pairing nondegenerate at analytic order two for elliptic curves, constructible from data consulting neither rank nor Selmer group, with an exhibited obstruction to such a pairing resolving the beachhead downward. Its confirmation threshold is curve-by-curve exhibition on a named curve of analytic order two plus a uniform construction, exactly as Section 5 states; its kill-switch is named; and the framework’s own machine-checked machinery is the instrument that would certify either outcome.

The call to the community is twofold and concrete. Arithmetic geometers: the order-two construction is the named deed, and the framework specifies its provenance conditions exactly enough to fail. Every field carrying a blocked row: when the next crossing arrives, execute the retro-type audit of Section 4.6 on it, and test whether the entering certificate was non-denotable in the prior even register, as P2 requires.

The single most important open question is the bridge question: does any analytic row, invariant subspace, Hilbert’s sixteenth, Lindelöf, admit an equivariant coordinate bridge into a finite frame at all, or is the wall at the continuous boundary absolute? The reframing acceptance would entail is one sentence: an open problem is not a hard problem but a blocked corpus missing one bit, and the whole craft of resolution is the dating of its entry.

AUTHOR’S PROVENANCE AND METHOD DISCLOSURE

This paper was developed under Trisduction, a verification and organizing discipline that adds the cited results no warrant; the method and its executable batteries are stated in full at the reference below.

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